Introduction to Quantum computing GDC 28/2/2025

$U|x\rangle$ if U is unitary

$\|x\| = \langle x|x\rangle^2 = 1 \implies 1 = \|U|x\rangle\| = \langle x|U^\dagger U x\rangle$

dagger

$U^\dagger = U^H$ $\quad U = [c_0 | c_1 \dots | c_{n-1}]$

① $\langle c_i|c_i\rangle = 1$

② $\langle c_i|c_j\rangle = 0$ columns are orthogonal

③ 1+2 the of U form an orthonormal basis

④ similarly also the rows of U are orthonormal

⑤ $U^{-1} = U^\dagger$

$(U^{-1})_{ij} = (\overline{u_{ji}})$

$$U^\dagger U = \begin{bmatrix} \langle c_0| \\ \vdots \\ \langle c_{n-1}| \end{bmatrix} \begin{bmatrix} |c_0\rangle & \dots & |c_{n-1}\rangle \end{bmatrix} = \begin{bmatrix} \langle c_0|c_0\rangle & \langle c_0|c_1\rangle & \dots & \langle c_0|c_{n-1}\rangle \\ & & \langle c_i|c_j\rangle & \end{bmatrix} = I$$

$H^2 = I$ $\quad H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$|\psi_0\rangle \xrightarrow{U_0} |\psi_1\rangle \xrightarrow{U_1} |\psi_2\rangle - - - \quad |\psi_F\rangle$

initial state

Evolution of quantum systems is unitary

$|\psi_k\rangle = U_{k-1}|\psi_{k-1}\rangle$ $\quad U_{k-1}$ unitary

$A^\dagger = A$ Hermitian matrices

$A^T = A$ symmetric matrices over the reals

Normal matrices: $A^\dagger A = A A^\dagger$

- symmetric $A^T = A$ $\quad A^T A = A^2 = A A^T$
- hermitian
- unitary $U^\dagger U = I = U U^\dagger$

$\vdots$

Theorem (spectral theorem)

A is normal $\Leftrightarrow$ $\exists$ P unitary such that $A = P D P^\dagger$ where D is diagonal, and $D = \begin{bmatrix} \lambda_0 & & \\ & \ddots & \\ & & \lambda_{n-1} \end{bmatrix}$

$A|P_i\rangle = \lambda_i |P_i\rangle$

$P = \big[\, |P_0\rangle \,\big|\, |P_1\rangle \,\big|\, \cdots \,\big|\, |P_{n-1}\rangle \,\big]$

$\{|P_i\rangle\}_{i=0\ldots n-1}$ is an ONB

Example

$\sigma_X = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\qquad X^\dagger X = X^2 = I$

$X = H \underbrace{D}_{\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}} H$ $\qquad$ eigen decomposition of X

$X \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \underset{\lambda_0}{1} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ $\qquad X \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = -1 \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

$$\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = |+\rangle = \frac{1}{\sqrt{2}}\underbrace{\begin{bmatrix}1\\0\end{bmatrix}}_{|0\rangle} + \frac{1}{\sqrt{2}}\underbrace{\begin{bmatrix}0\\1\end{bmatrix}}_{|1\rangle} = \frac{(|0\rangle + |1\rangle)}{\sqrt{2}}$$

$$\frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix} = |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$\boxed{A = \sum_{j=0}^{n-1} \lambda_j \cdot \underbrace{|c_j\rangle\langle c_j|}_{\text{rank-1 matrix}}}$$

$$X = \lambda_0 \; |+\rangle\langle+| + \lambda_1 |-\rangle\langle-|$$

$$= 1 \, |+\rangle\langle+| - |-\rangle\langle-|$$

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$$

$$\frac{1}{2}\begin{bmatrix}1\\-1\end{bmatrix}\begin{bmatrix}1 & -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix} - \frac{1}{2}\begin{bmatrix}1 & -1\\-1 & 1\end{bmatrix} = \begin{bmatrix}0 & \frac{1}{2}+\frac{1}{2}\\ \frac{1}{2}+\frac{1}{2} & 0\end{bmatrix} = X$$

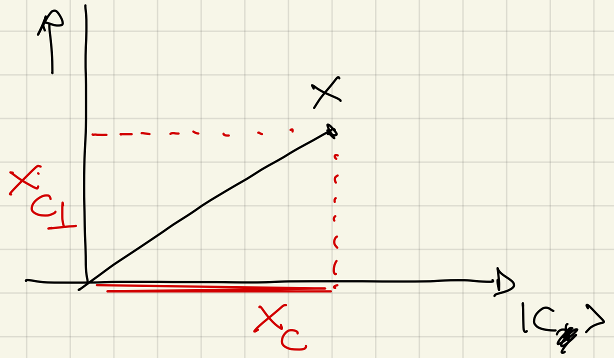
$P = |c_i\rangle\langle c_i|$ is an orthogonal projector onto the eigenspace of λ_i

Def P is a projector if $P^2 = P$

a projector is called orthogonal $P^\dagger = P$

$$P^2 = |c_i\rangle\langle c_i| \, |c_i\rangle\langle c_i| = \underbrace{\langle c_i | c_i\rangle}_{1} |c_i\rangle\langle c_i| = P$$

$$P^\dagger = \left(|c_i\rangle\langle c_i|\right)^\dagger = |c_i\rangle\langle c_i| = P$$



$P = |c\rangle\langle c|$

$$x = x_c + x_{c\perp}$$

$$P|x\rangle = |c\rangle\langle c| \, |x\rangle = \left(|c\rangle\langle c|\right)\left(|x_c\rangle + |x_{c\perp}\rangle\right) =$$

$$= |c\rangle\langle c | x_c\rangle + \underbrace{|c\rangle\langle c | x_{c\perp}\rangle}_{=0}$$

$$= \langle c | x_c\rangle \, |c\rangle$$

$$\left(|c\rangle\langle c|\right)^m = |c\rangle\underbrace{\langle c|c\rangle}_{1}\underbrace{\langle c|c\rangle}_{1}\langle c| - \ldots \, |c\rangle\langle c| = |c\rangle\langle c|$$

$$P^m = P$$

$$A^m = \sum_{j=0}^{n-1} \lambda_j^m \, |c_j\rangle\langle c_j|$$

$$A^m = \overline{P D \underbrace{P^+ P}_{I} D P^+ D \ldots \quad P^+} = P D^m P^+$$

$$f(x) = \sum_{m\geq 0}^{\infty} a_m x^m$$ power series of a function f

$$f(x) = e^x = \sum_{m\geq 0}^{\infty} \frac{1}{m!} x^m$$

is a matrix

$$\boxed{f(A)} = \sum_{m\geq 0}^{\infty} a_m A^m$$

$$A \in \mathbb{C}^{n\times n}$$

$$e^A = \sum_{m\geq 0}^{\infty} \frac{1}{m!} A^m = \sum_{m\geq 0}^{\infty} \frac{1}{m!} \left(\sum_{j=0}^{n-1} \lambda_j^m \, |c_j\rangle\langle c_j| \right)$$

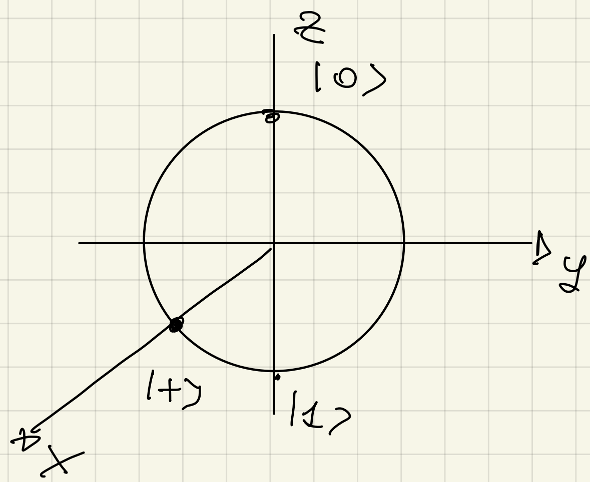
$$= \sum_{j=0}^{n-1} \underbrace{\left(\sum_{m\geq 0}^{\infty} \frac{\lambda_j^m}{m!} \right)}_{= e^{\lambda_j}} |c_j\rangle\langle c_j|$$

$$= P \begin{bmatrix} e^{\lambda_0} & & & \\ & e^{\lambda_1} & & \\ & & \ddots & \\ & & & e^{\lambda_{n-1}} \end{bmatrix} P^+$$

$$f(A) = P \begin{bmatrix} f(\lambda_0) & & \\ & \ddots & \\ & & f(\lambda_{n-1}) \end{bmatrix} P^\dagger = \sum_{j=0}^{n-1} f(\lambda_j)|g_j\rangle\langle g_j|$$

Exercise : $A^\dagger = A$ A hermetian $\Rightarrow$ e^A is unetory

$$\sigma_x = X|+\rangle = |+\rangle$$



$$R_x(\vartheta) = e^{-i\vartheta/2\,\sigma_x}$$

$$\sigma_x = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = H \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} H$$

$$e^{-i\frac{\vartheta}{2}\sigma_x} = H \begin{bmatrix} e^{-i\vartheta/2 \cdot \lambda_0^{''1}} & \\ & e^{-i\vartheta/2\, \lambda_1^{''-1}} \end{bmatrix} H$$

$$= H \begin{bmatrix} e^{-i\vartheta/2} & 0 \\ 0 & e^{i\vartheta/2} \end{bmatrix} H = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{-i\vartheta/2} & \\ & e^{i\vartheta/2} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\cos\vartheta/2 - i\sin\frac{\vartheta}{2}$

$\cos\frac{\vartheta}{2} + i\sin\vartheta/2$

$$= \begin{bmatrix} \cos\vartheta/2 & -i\sin\vartheta/2 \\ -i\sin\frac{\vartheta}{2} & \cos\vartheta/2 \end{bmatrix} = R_x(\vartheta)$$

$$R_y(\vartheta) = e^{-i\vartheta/2\,\sigma_y} \qquad \sigma_y = Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} e^{-i\vartheta/2} & 0 \\ 0 & e^{i\vartheta/2} \end{bmatrix} = e^{-i\vartheta/2} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\vartheta} \end{bmatrix}$$

Rotation of an arbitrary angle about axis z

$$R_z(\vartheta) = e^{-i\vartheta/2\,\sigma_z} \qquad \sigma_z = Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\frac{\vartheta}{2} & \sin\frac{\vartheta}{2} \\ -\sin\frac{\vartheta}{2} & \cos\frac{\vartheta}{2} \end{bmatrix}$$

$$\boxed{\begin{gathered} A : \quad A^2 = I \\ e^{iAx} = \cos x \, I + i \sin x \, A \end{gathered}}$$

Theorem: Any 2×2 unitary matrix U is such that there exist $\alpha, \beta, \gamma, \delta \in \mathbb{R}$ such that

$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$$

Similar decomposition using other pauli matrices

Multiqubit systems

$H_1 \otimes H_2$

$\dim H_1 = n$

$\dim H_2 = m$

$\dim(H_1 \otimes H_2) = n \cdot m$

$\{|b_i\rangle\}_{i=0 \ldots n-1}$ basis for H_1

$\{|c_j\rangle\}_{j=0 \ldots m-1}$ basis for H_2

$\{|b_i\rangle \otimes |c_j\rangle\}_{\substack{i=0 \ldots n-1 \\ j=0 \ldots m-1}}$

$$|\varphi\rangle = \begin{bmatrix} \varphi_0 \\ \varphi_1 \end{bmatrix} \qquad |\psi\rangle = \begin{bmatrix} \psi_0 \\ \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix}$$

kronecker

$$|\varphi\rangle \otimes |\psi\rangle = \begin{bmatrix} \varphi_0 \psi_0 \\ \varphi_0 \psi_1 \\ \varphi_0 \psi_2 \\ \varphi_0 \psi_3 \\ \varphi_1 \psi_0 \\ \varphi_1 \psi_1 \\ \varphi_1 \psi_2 \\ \varphi_1 \psi_3 \end{bmatrix}$$

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \dots & a_{1n}B \\ \vdots & & & \\ a_{n1}B & a_{n2}B & \dots & a_{nn}B \end{bmatrix} nm$$

$\underset{n \times n}{A} \quad \underset{m \times m}{B}$

nm

Axiomatic characterization of $|\varphi\rangle \otimes |\psi\rangle$

1. $c(|\varphi\rangle \otimes |\psi\rangle) = |c\varphi\rangle \otimes |\psi\rangle = |\varphi\rangle \otimes |c\psi\rangle$

2. $(|\psi_1\rangle + |\psi_2\rangle) \otimes |\varphi\rangle = |\psi_1\rangle \otimes |\varphi\rangle + |\psi_2\rangle \otimes |\varphi\rangle$

3. $(A \otimes B)(|\varphi\rangle \otimes |\psi\rangle) = A|\varphi\rangle \otimes B|\psi\rangle$

$$|\varphi\rangle \otimes |\psi\rangle = |\varphi\rangle|\psi\rangle = |\varphi\psi\rangle$$

$$|0\rangle \otimes |1\rangle = |0\rangle|1\rangle = \underline{\underline{|01\rangle}}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \leftarrow$$

$$|11\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$|0\rangle \ |1\rangle \ \dots \ |N-1\rangle$ $\qquad N = 2^n$

$|$ $\quad |$ $\quad |$

$N = 16 \qquad n = 4$

$|0000\rangle \quad |0001\rangle \quad \dots \quad |1111\rangle$

Schmidt decomposition theorem

$|\psi\rangle \in H_A \otimes H_B$

There exist ONB $\{|\varphi_i^A\rangle\}$, and $\{|\varphi_j^B\rangle\}$
and $p_i \geq 0$
such that

$$|\varphi\rangle = \sum_{i=0}^{\min(\dim H_A, \dim H_B)} \sqrt{p_i}\, |\varphi_i^A\rangle |\varphi_i^B\rangle$$